Estimating Production Functions in Markets with Strategic Interactions: an MPEC Approach*

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Abstract

Standard production function estimators assume firms are price-takers in output markets, yet many industries are characterized by oligopolistic competition. Under imperfect competition, the firm's input demand depends on the demand shock and competitors' actions, violating the scalar unobservability condition that underpins control function approaches. This paper develops an MPEC-GMM estimator that retains all latent variables as explicit optimization variables and enforces the model's structural equations as equality constraints, jointly recovering production function parameters, demand parameters, and firm-level productivity without analytical inversion. I derive the estimator under both Cournot and Bertrand competition, showing that the two modes share a common estimation structure with only the first order condition and demand specification differing. I further show that specifying demand as a function of observed rather than latent output enables a computationally advantageous two-stage procedure. Monte Carlo simulations for the Cobb-Douglas case with linear demand under Cournot competition confirm that the estimator recovers all structural parameters, with output elasticities and demand parameters estimated precisely and all confidence intervals covering the true values.

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1 Introduction

The estimation of production functions is central to empirical economic research in a variety of fields such as industrial organization, economics of trade, or labor economics. Despite its importance, robustly estimating its parameters under realistic market assumptions is still a largely unsolved issue.

Since the seminal contributions of Olley and Pakes (1996), Levinsohn and Petrin (2003), and Ackerberg et al. (2015) the proxy variable approach and the dynamic panel methods of Blundell and Bond (2000) have become workhorses in applied research. However, a maintained assumption in almost all production function estimation approaches is that firms take output prices as given and do not take their competitors' choices into consideration when making their production decisions. However, this restrictive assumption is clearly violated in many industries of interest such as pharmaceuticals, automobiles, cement, beer, and most manufacturing sectors. In most markets at least some firms possess market power, face downward-sloping demand curves, and take competitors' choices into account. Ignoring these aspects can lead to substantially biased estimates of the production function parameters (Jaumandreu, 2025; Ackerberg and De Loecker, 2024; Doraszelski and Li, 2025).

The central challenge posed by imperfect competition for proxy variable production function estimation approaches stems from the behavioral restrictions they impose on the proxy variable. The proxy variable estimation approach relies on inverting the firm's demand function to recover unobserved productivity. However, under imperfect competition, the input demand depends not only on own productivity but also on competitors' actions and demand shocks, which are often unobserved. Not taking these quantities into account violates the scalar unobservable assumption required for inversion and leads to a bias in the estimates (Ackerberg and De Loecker, 2024; Jaumandreu, 2025).

In this paper, I develop a general production function estimation approach that takes strategic interactions in the output market into account. The approach can be applied to estimating parameters of any commonly used production function type, under both Bertrand or Cournot competition, and any commonly employed demand formulation.[1] Instead of requiring a parametric or nonparametric inversion of an input demand function, my approach builds on insights from Trunschke and Judd (2024), whose approach uses first order conditions to express unobserved productivity as a known function of observables and parameters. In the approach I develop in this paper, I formulate the estimation problem in a mathematical programming with equilibrium constraints (MPEC) framework (Su and Judd, 2012). The approach embeds the relevant structural relationships originating from the firm's profit maximization problem in a system of constraints to create a single-stage MPEC-GMM estimator. Instead of concentrating out latent variables, the constraints encode the firm's first-order conditions for profit maximization under imperfect competition together with the specified production function, demand system, and the law of motion for productivity.

This approach has several advantages over existing estimators. First, it avoids the need to nonparametrically invert the input demand equation to recover productivity entirely, sidestepping the invertibility problems highlighted by Doraszelski and Li (2025). By treating productivity $\Omega_{j,t}$ as an explicit optimization variable, the MPEC approach is immune to the failures of scalar unobservability that plague proxy variable methods under imperfect competition. Second, unlike the dynamic panel approach, which requires a linear Markov process for productivity, the MPEC approach naturally accommodates nonlinear productivity dynamics. Third, my approach can accommodate flexible demand

[1] If data on product characteristics is available, the estimator can be combined with a discrete choice demand system in characteristics space. However, this paper focuses on the situation where only information on firms' input and output decisions, and prices is available.

systems such as the Almost Ideal Demand System (Deaton and Muellbauer, 1980), while ensuring internal consistency between the production function specification, demand system, and optimality conditions, which is difficult to guarantee in semiparametric two-step procedures. Furthermore, the MPEC-GMM approach is a one-step estimator making estimation generally more efficient than two-step procedures while allowing to use standard asymptotic variance expressions. Several recent papers have proposed alternative solutions within existing frameworks, including extended proxy variable inversions, and cost-share estimators robust to market power.

In this paper, I first derive the MPEC-GMM estimator in a general framework for both Cournot and Bertrand competition, allowing for arbitrary production functions and general demand systems. This generalizes the FOC-based constrained GMM approach of Trunschke and Judd (2024) where the first order constraint directly solves for productivity from perfect to imperfect competition. I derive the full MPEC estimator for both competition modes, showing that the Cournot and Bertrand cases share the same constraint structure but differ in the relevant demand object entering the first-order condition. Second, I analyze an important specification choice: whether to specify the demand function in terms of unobserved output or observed output, where the observed output includes an ex-post error term. I show that using Y has the advantage that demand can be estimated in a first stage using only observables, which pins down the demand shock δ_t and reduces the dimensionality of the second-stage problem. However, it changes the firm's first-order conditions in a non-trivial way as the firm maximizes expected profit rather than profit directly, introducing two additional parameters into the first order constraint. When additionally assuming normality of the error term, only one additional parameter needs to be estimated. Third, I derive the specific GMM-MPEC estimator from the general framework in the case of a Cobb-Douglas production function with linear demand under Cournot competition, and provide Monte Carlo evidence that the estimator can recover the true parameter values even in relatively small samples.

The remainder of the paper is organized as follows. Section 2 presents the general model and derives the MPEC estimators for both Cournot and Bertrand competition. Section 3 analyzes the design choice between observed demand and unobserved demand, and derives the modified first-order conditions under each. Section 4 provides a detailed comparison with other approaches in the literature. Section 5 specializes the framework to homogeneous-goods Cournot competition with Cobb-Douglas production and linear demand, and presents Monte Carlo evidence of the estimator's performance. Section 6 concludes.

2 General Approach

A production function characterizes the technology by which a firm transforms production inputs into output with common specifications such as the Cobb–Douglas, constant elasticity of substitution (CES), and Translog. Irrespective of the particular functional form, production functions typically contain a term that captures the firm's efficiency in transforming inputs into output, commonly referred to as Total Factor Productivity (TFP). Following the standard approach in the literature, I assume TFP to be Hicks-neutral, meaning that changes in productivity do not affect the relative proportions of inputs used in production.[2] Assumption 1 states this condition formally.

[2] A strand of the literature relaxes this assumption, typically by introducing additional productivity components that influence only a subset of inputs (Doraszelski and Jaumandreu, 2018; Raval, 2019).

Assumption 1 *Each period t, firm j chooses production output $Q_{j,t}$ using capital $K_{j,t}$, labor $L_{j,t}$, and intermediate inputs $M_{j,t}$ and the production function*

$$F(K_{j,t}, L_{j,t}, M_{j,t}; \beta), \tag{1}$$

such that

$$Q_{j,t} = F(K_{j,t}, L_{j,t}, M_{j,t}; \beta)\Omega_{j,t}, \tag{2}$$

where $F(K_{j,t}, L_{j,t}, M_{j,t}; \beta)$ is at least once differentiable in $M_{j,t}$. β represents a vector of parameters describing output and possibly substitution elasticities. $\Omega_{j,t}$ describes firm i's TFP in period t. Output $Q_{j,t}$ is unobservable to the econometrician but

$$Y_{j,t} = Q_{j,t}e^{\epsilon_{j,t}}, \tag{3}$$

is observable, where $\epsilon_{j,t}$ is an i.i.d. error term, with $E[\epsilon_{j,t}] = 0$.

Firms observe their TFP, $\Omega_{j,t}$, before they make decisions on output $Q_{j,t}$ and incorporate this information. Typically TFP, $\Omega_{j,t}$, is not included in firm data and ignoring the connection between firms' decisions and TFP when estimating production function parameters causes the well-known transmission bias [3].

As is common in the literature, I allow total factor productivity $\Omega_{j,t}$ to differ across firms and over time and assume it to be imperfectly persistent. Specifically, log productivity $\omega_{j,t} = \ln \Omega_{j,t}$ follows a first-order Markov process $g(\omega_{j,t-1}, \rho_\omega)$ that includes an exogenous i.i.d. productivity shock $\xi_{j,t}$ satisfying the mean independence condition $E[\xi_{j,t}|\omega_{j,t-1}] = 0$. This conditional mean independence restriction is central to identification: it implies that the productivity shock is orthogonal to lagged productivity and, by extension, to all input choices made before its realization, providing the key moment conditions for estimation. Neither the log-productivity level $\omega_{j,t}$ nor the productivity shock $\xi_{j,t}$ are observed in the data, but both are known to the firm at the time it makes its flexible input and output decisions. Assumption 2 formalizes this, with lowercase letters representing logarithms of their uppercase counterparts throughout.[4]

Assumption 2 *$\omega_{j,t}$ follows a first-order Markov process*

$$\omega_{j,t} = g(\omega_{j,t-1}, \rho_\omega) + \xi_{j,t}, \text{ with } E[\xi_{j,t}|\omega_{j,t-1}] = 0, \tag{4}$$

where $\xi_{j,t}$ represents a random exogenous productivity shock for firm j in period t. Neither $\omega_{j,t}$ nor $\xi_{j,t}$ are observed in the data, but they are known to the firm.

Production inputs are typically classified as either *flexible* or *fixed*, depending on their timing relative to the realization of the productivity shock $\xi_{j,t}$. This distinction is crucial for identification, as it determines which inputs are predetermined with respect to $\xi_{j,t}$ and therefore provide valid moment conditions for estimation. I define a *flexible input* as any input the firm can freely choose after observing its contemporaneous productivity level $\Omega_{j,t}$ and whose choice is realized within the same period. Note

[3] see Levinsohn and Petrin (2003) for a derivation of the bias in the Cobb-Douglas case

[4] This assumption can be weakened by allowing $g(\cdot)$ to include additional lagged values of $\omega_{j,t}$ or other variables. However, I use the simple first-order Markov process because most of the methodological and applied literature does not employ higher orders. This is without loss of generality of the identification arguments.

that flexible inputs' optimal levels depend on the current realization of productivity and do not depend on their own past values. Examples of inputs likely to satisfy these conditions include intermediate inputs such as materials and energy. A *fixed input*, by contrast, is any input that the firm cannot freely adjust in response to the contemporaneous productivity shock. This category includes two distinct cases: inputs chosen within the current period but before the realization of $\xi_{j,t}$, such as labor, and inputs whose levels are determined by decisions made in prior periods that only materialize in the current one, such as physical capital or machinery.[5] Because fixed inputs are chosen without knowledge of the contemporaneous productivity shock $\xi_{j,t}$, they are orthogonal to it by construction, which is what makes them valid instruments. The flexible input, on the other hand, responds to $\xi_{j,t}$ and therefore correlates with unobserved productivity, but this responsiveness simplifies its first order condition in the profit maximization problem making an ideal model restriction for the estimation approach. As common in the literature, I assume intermediate inputs to be fully flexible, labor to be fixed and chosen in the current period before observing the firm's current productivity level, and capital to be fixed and predetermined by investment decisions made in the previous period (Gandhi et al., 2020) . Assumption 3 formally summarizes the input types and their choice timings.

Assumption 3 *Input decisions timing and their materializations are as follows:*

(a) ***Capital*** *(fixed, predetermined): Capital input $K_{j,t}$ is determined by the firm's investment decision $I_{j,t-1}$ made in the previous period and is therefore fixed throughout period t. Capital depreciates at some rate between periods. Because investment decisions are made one period in advance, capital is predetermined with respect to both the contemporaneous productivity level $\Omega_{j,t}$ and the productivity shock $\xi_{j,t}$.*

(b) ***Labor*** *(fixed, contemporaneous): Labor input $L_{j,t}$ is chosen in period t and realized instantly, but the firm commits to its labor level before observing the contemporaneous productivity shock $\xi_{j,t}$. Labor therefore depends on the expected productivity $E[\omega_{j,t}|\omega_{j,t-1}]$ but not on the realized productivity shock.*

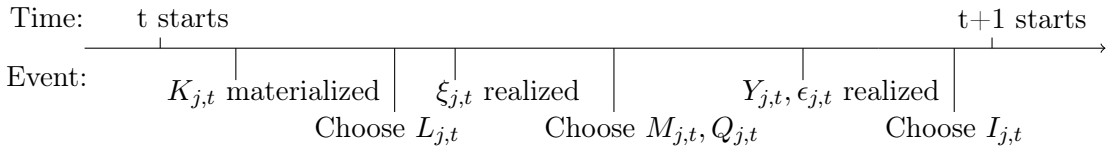
(c) ***Intermediate inputs*** *(flexible): Intermediate inputs $M_{j,t}$ are chosen after the firm observes its full contemporaneous productivity $\omega_{j,t}$, including the productivity shock $\xi_{j,t}$, and are realized within the same period.*

Assumption 3 induces the sequential decision timing within each period, shown in Figure 1. At the beginning of period t, the firm's capital stock $K_{j,t}$ is already determined by its investment decision $I_{j,t-1}$ from the previous period. Given this predetermined capital level, the firm chooses its labor input $L_{j,t}$ based on the conditional expectation $E[\omega_{j,t}|\omega_{j,t-1}]$ but before observing the contemporaneous productivity shock $\xi_{j,t}$. As a consequence, labor depends on the previous period's productivity level and the expected productivity transition but not on the realized shock, which is what generates its orthogonality to $\xi_{j,t}$. Once the productivity shock $\xi_{j,t}$ is realized and observed by the firm, the firm simultaneously chooses its intermediate input level $M_{j,t}$ and output quantity $Q_{j,t}$, both of which therefore respond to the contemporaneous productivity $\omega_{j,t}$. Finally, the firm makes its investment decision $I_{j,t}$ for the subsequent period's capital stock. It is not necessary to specify the capital investment process further for the estimation approach described below as it makes no additional assumptions on

[5]Depending on the context, assuming labor to be flexible can be valid. However, labor adjustments can often take time as labor protection laws regulate hiring and layoffs. The estimation approach can be generalized to a situation with multiple flexible inputs; even though it is not explicitly derived in this paper.

capital accumulation or optimality of capital choices. This timing structure implies a strict ordering of the econometrician's instruments: capital, being determined in the previous period before the productivity shock it must be orthogonal to, provides the strongest exclusion restriction, followed by labor, which is determined within the same period but before knowledge of the productivity shock.

Figure 1: Decision Timing



I summarize all relevant information the firm possesses at the time it makes its flexible input and output choices in an information set $\mathbb{I}_{j,t}$, formally defined in Assumption 4. This set plays a central role in constructing moment conditions: any variable that is in $\mathbb{I}_{j,t}$ or determined prior to it is a candidate instrument, because the firm's choices depend on all information within this set.

Assumption 4 *Set $\mathbb{I}_{j,t}$ describes firm j's information set in period t after the productivity shock $\xi_{j,t}$ is realized. It includes all previous realizations of the firm's state variables, but it does not contain any information about future realizations of any variable.*

Crucially, the contemporaneous productivity level $\omega_{j,t}$ and the productivity shock $\xi_{j,t}$ are elements of $\mathbb{I}_{j,t}$, so the firm's flexible input and output decisions are functions of them. The measurement error $\epsilon_{j,t}$, by contrast, is not in $\mathbb{I}_{j,t}$ as it is realized after all input and output decisions are made and is therefore orthogonal to every element of the information set by construction.

The remaining assumptions concern the firm's objective and its market power. I assume that firms maximize contemporaneous profits when choosing their flexible input $M_{j,t}$ (Assumption 5). Because capital and labor are fixed at the time of this decision, they enter the information set $\mathbb{I}_{j,t}$ as predetermined state variables: the firm chooses $M_{j,t}$ and $Q_{j,t}$ conditional on $K_{j,t}$ and $L_{j,t}$. This means that the profit maximization problem is static in the flexible input. The firm takes its fixed inputs as given and optimizes over intermediate inputs and output only. The resulting first order condition for $M_{j,t}$ provides a key structural restriction that links observable input prices to unobservable productivity and output through the production function and, under imperfect competition, the demand system. I assume that the econometrician observes output prices and flexible input prices for each firm in each period.[6]

Assumption 5 *Firms choose their flexible production input $M_{j,t}$ to maximize contemporaneous profits $\Pi_{j,t}$, conditional on $\mathbb{I}_{j,t}$,*

$$\Pi_{j,t} = P^Q_{j,t} Q_{j,t} - P^M_{j,t} M_{j,t}, \tag{5}$$

where $P^Q_{j,t}$ denotes the price of firm j's output and $P^M_{j,t}$ denotes the price of intermediate inputs in period t.

[6]If prices are not available, they can be replaced with industry input and output price indices. However, this substitution is only valid in settings where the imperfect competition game leads to a common price across firms. This heavily restricts the choice of the demand system and the mode of competition. For example, in the homogeneous goods case under Cournot competition, the market price is the same for all firms, so using an industry price index is valid. However, under virtually all heterogenous goods demand systems, firms will generally set different prices making the substitution invalid.

Assumptions 6 and 7 specify the firm's market power on input and output markets, respectively.

Assumption 6 *Firms are price-takers in input markets: each firm takes the intermediate input price $P^M_{j,t}$ as exogenously given and cannot influence it through its input choices.*

Assumption 7 *Firms face downward-sloping demand in output markets and choose their output levels to maximize contemporaneous profits, recognizing the effect of their decisions on the output price $P^Q_{j,t}$.*

Assumption 6 is necessary because if firms could influence input prices, their input decisions would depend on price elasticities in input markets and potentially on competitors' input choices, substantially complicating the first order conditions. It also ensures that input prices are exogenous to the demand shock, making them valid instruments for demand estimation. Most approaches in the literature additionally assume firms to be price-takers in the output market, primarily for simplicity and because it avoids the need to specify a demand system. This paper relaxes the output market assumption to Assumption 7, allowing firms to exercise market power without requiring additional data beyond what standard approaches use.

This paper focuses on settings in which firms compete strategically in output markets, taking their competitors' actions into account when making production decisions. The combination of market power with unobserved productivity creates a joint identification problem: the firm's first order condition for the flexible input now involves both the production function parameters β and the demand parameters η, because the markup depends on the slope of the demand curve. Under Assumptions 1-5 production function and demand parameters must therefore be estimated simultaneously rather than sequentially. I derive the estimator under two canonical modes of oligopolistic competition: Cournot competition, in which firms set the quantity of their good and the market price adjusts through an inverse demand function (Section 2.1), and Bertrand competition, in which firms set output prices and quantities are determined by the demand system (Section 2.2). Although the two modes differ in the firm's strategic variable and the relevant demand derivative entering the first order condition, the overall estimation structure, an MPEC that jointly recovers production function parameters, demand parameters, and latent productivities, is common to both.

2.1 Cournot Case

Under Cournot competition, each firm simultaneously chooses its output quantity taking competitors' quantities as given, with the market price determined by the inverse demand function.

Assumption 8a *Firms compete in quantities. Firms act non-cooperatively and each firm simultaneously chooses its output level taking the output choices of all other firms as given.*

Under Assumptions 1–8a, I focus on Nash equilibria in which each firm's quantity choice is a best response to the quantities chosen by all other firms. The firm's choice variables are the intermediate input $M_{j,t}$ and output quantity $Q_{j,t}$, linked through the production function; capital and labor are predetermined and enter as state variables. The output price $P^Q_{j,t}$ is not a choice variable of the firm but is determined endogenously through the inverse demand function D^C, which maps the full vector of market quantities $Q_t = (Q_{1,t}, \ldots, Q_{J,t})'$ and a common demand shock δ_t to a vector of market prices.

Firm j's profit maximization problem in period t is

$$\begin{aligned} \max_{M_{j,t},Q_{j,t}} \quad & \pi_{j,t} = P_{j,t}^{Q} Q_{j,t} - P_{j,t}^{M} M_{j,t}, \\ \text{s.t.} \quad & P_{j,t}^{Q} = D^{C}(Q_t, \delta_t; \eta), \\ & Q_{j,t} = F(K_{j,t}, L_{j,t}, M_{j,t}; \beta)\Omega_{j,t}. \end{aligned} \tag{6}$$

Because the firm takes competitors' quantities as given, the derivative of revenue with respect to its own output involves both the direct revenue from selling an additional unit at price $P_{j,t}^{Q}$ and the revenue loss from lowering the market price through inverse demand. Differentiating profits with respect to $M_{j,t}$ and applying the chain rule through the production function yields the first order condition (FOC):

$$\frac{\partial \pi(K_{j,t}, L_{j,t}, M_{j,t}, \Omega_{j,t}, P_{j,t}^{Q}, P_{j,t}^{M}; \beta, \eta)}{\partial M_{j,t}} = \frac{\partial P_{j,t}^{Q}}{\partial Q_{j,t}} \frac{\partial Q_{j,t}}{\partial M_{j,t}} Q_{j,t} + \frac{\partial Q_{j,t}}{\partial M_{j,t}} P_{j,t}^{Q} - P_{j,t}^{M} = 0 \tag{7}$$

The first term captures the markup adjustment. Expanding output reduces the market price on all inframarginal units, so the firm internalizes its effect on revenue through inverse demand. Under perfect competition this term vanishes and the FOC reduces to the standard condition that output price equals marginal cost as in Trunschke and Judd (2024). This FOC holds generally under any differentiable production function and demand specification. Depending on the market, the demand function can be highly dimensional to accommodate heterogeneous goods or relatively parsimonious as in the homogeneous goods case.

As in the perfect competition case of Trunschke and Judd (2024), this FOC can serve as a constraint in the estimation problem. However, imperfect competition introduces fundamental complications. The FOC now contains the partial derivative of the inverse demand function $\frac{\partial P_{j,t}^{Q}}{\partial Q_{j,t}}$, which makes it necessary to specify and jointly estimate a demand function. Moreover, because inverse demand maps the entire vector of market quantities Q_t to a price, this derivative is generally a function of all firms' latent output quantities and, through the production function, of their latent productivities. This creates a system of interdependent FOCs across firms within the same market. Even in the special case where the FOC depends only on firm j's own quantity and productivity, inverting it to solve for $\Omega_{j,t}$ analytically, as is possible under perfect competition in Trunschke and Judd (2024), is not generally feasible. The FOC does not need to be injective in the unobserved quantities for a given set of parameters and observed variables. I show that even in the simplest case of linear demand with homogeneous goods, the FOC is quadratic in $Q_{j,t}$, yielding two positive solutions that both satisfy the second order conditions. The MPEC formulation resolves this difficulty. Rather than inverting the FOC analytically, the solver jointly recovers the full vectors of output quantities Q_t and productivities Ω_t that simultaneously satisfy all firms' FOCs as constraints while minimizing the GMM objective function.

Constructing the MPEC requires enough constraints to pin down all unobserved variables. For each firm in each period, there are four latent variables, output quantity $Q_{j,t}$, productivity $\Omega_{j,t}$, measurement error $\epsilon_{j,t}$, and the productivity shock $\xi_{j,t}$, plus one common demand shock δ_t per period. The FOCs alone provide one constraint per firm-period, which is insufficient. However, the model's structural equations supply the remaining restrictions. The assumed relationship in Equation (3) links observed output $Y_{j,t}$ to the firm's output choice $Q_{j,t}$ and the measurement error $\epsilon_{j,t}$, providing one constraint per firm-period. The production function (Assumption 1) ties $Q_{j,t}$ to inputs and $\Omega_{j,t}$, providing another. The productivity development process (Assumption 2) relates $\xi_{j,t}$ to $\omega_{j,t}$ and $\omega_{j,t-1}$, adding one more

per firm-period. Finally, the inverse demand function provides T additional constraints linking the market price to the vector of quantities and the demand shock δ_t. Together, these constraints match the number of latent variables.

Identification additionally requires at least as many moment conditions as parameters, and preferably more for overidentification. Following the literature (Ackerberg et al., 2015), the moment conditions exploit the information set structure formalized in Assumption 4. The first set of moments arises from the orthogonality of the productivity shock $\xi_{j,t}$ to predetermined variables. Because capital and labor are chosen before $\xi_{j,t}$ is realized (Assumption 3), they are valid instruments and $E[\xi_{j,t}|K_{j,t}] = 0$, $E[\xi_{j,t}|L_{j,t}] = 0$ hold. Lags of all inputs, including intermediate inputs, are also orthogonal to the contemporaneous productivity shock $\xi_{j,t}$, providing additional moment conditions and can be used to form additional moment conditions. The Markov assumption on productivity (Assumption 2) implies that $\xi_{j,t}$ is orthogonal to the lagged productivity level and its powers, yielding further moment conditions. Because $\epsilon_{j,t}$ is not in $\mathbb{I}_{j,t}$, any contemporaneous or lagged input choice is orthogonal to it by construction, providing another block of moments. For the demand parameters, the contemporaneous demand shock δ_t is observed by the firm and therefore correlated with flexible input choices and output quantities, ruling these out as instruments. However, input prices serve as valid instruments for demand. Under Assumption 6, they are exogenous to the demand shock, yet they affect firms' output decisions through the profit maximization problem, satisfying both the exclusion restriction and the relevance condition. Finally, if the productivity process has reached its ergodic distribution in the sample, the unconditional mean of productivity provides an additional moment condition that helps pin down the scale coefficient of the production function.

Combining the structural constraints and moment conditions, I can state the full MPEC estimator under Cournot competition. Let $A(X, \sigma; \theta)$ denote the sample analogue of the stacked moment conditions, with X collecting all observed data, $\sigma = (Q_{j,t}, \Omega_{j,t}, \epsilon_{j,t}, \xi_{j,t}, \delta_t)$ collecting all latent variables, and let W denote an appropriate weighting matrix. The parameter vector $\theta = (\beta, \rho, \eta)$ collects all unknown parameters. The estimator jointly minimizes the GMM objective over θ and the latent variables σ, subject to all structural equations holding as equality constraints for every firm j and period t:

$$
\begin{aligned}
\underset{\theta,\sigma}{\arg\min} \quad & \frac{1}{J \cdot T} A(X;\theta)^T W A(X;\theta) \\
s.t. \quad & P^M_{j,t} = \frac{\partial P^Q_{j,t}}{\partial Q_{j,t}} \frac{\partial Q_{j,t}}{\partial M_{j,t}} Q_{j,t} + \frac{\partial Q_{j,t}}{\partial M_{j,t}} P^Q_{j,t} - P^M_{j,t}, \\
& \omega_{j,t} = g(\omega_{j,t-1};\rho) + \xi_{j,t}, \\
& Q_{j,t} = F(K_{j,t}, L_{j,t}, M_{j,t};\beta)\Omega_{j,t}, \\
& Y_{j,t} = Q_{j,t} e^{\epsilon_{j,t}}, \\
& P^Q_{j,t} = D^C(Q_t, \delta_t;\eta), \qquad \forall t = 1, ..., T, \\
& \qquad\qquad\qquad\qquad\quad \forall j = 1, ..., J.
\end{aligned}
\tag{8}
$$

2.2 Bertrand Case

I now turn to the case where firms compete by setting prices, known as Bertrand competition. Under this mode of competition, each firm simultaneously chooses its output price $P^Q_{j,t}$ taking all competitors' prices as given, and the demand system determines each firm's quantity sold. The key structural difference from the Cournot case is that prices rather than quantities are the strategic variables, which changes both the firm's optimization problem and the demand object entering the first order condition.

Assumption 8b *Firms compete in the output market by simultaneously setting the prices of their output. Firms act non-cooperatively. Marginal costs are strictly convex and the marginal cost of producing $Q_{j,t} = 0$ is zero.*

Assumption 8b addresses two issues that arise under Bertrand competition, particularly with homogeneous goods. First, strict convexity of costs, equivalently, strictly decreasing returns to scale in the production function, guarantees the existence of Nash equilibria even in the homogeneous goods case (Dastidar, 1995). Without this condition, in the homogeneous goods case, the classical Bertrand paradox applies and firms undercut each other until price equals marginal cost, potentially leaving no equilibrium when marginal costs are constant. Strict convexity also ensures that no single firm can profitably serve the entire market, so all firms produce positive quantities in equilibrium. Second, the condition that marginal cost at zero output is zero ensures that every firm strictly prefers producing to shutting down, ruling out corner solutions at $Q_{j,t} = 0$. Both conditions are automatically satisfied by standard production function specifications with decreasing returns to scale, including the Cobb-Douglas case with output elasticities summing to less than one.

Under Assumptions 1–8b, I focus on Nash equilibria in which each firm's price choice is a best response to the prices set by all other firms. The firm's choice variables are now the output price $P^Q_{j,t}$ and the intermediate input $M_{j,t}$. Unlike in the Cournot case, the demand function D^B_j maps the full vector of all firms' output prices $P^Q_t = (P^Q_{1,t}, \ldots, P^Q_{J,t})'$ and the demand shock δ_t to firm j's individual quantity demanded, rather than mapping a vector of quantities to a vector of market prices. This firm-specific demand structure naturally accommodates product differentiation, where each firm's quantity depends on its own price and on competitors' prices through cross-price elasticities. Firm j's profit maximization problem in period t is

$$\begin{aligned} \max_{M_{j,t}, P^Q_{j,t}} \quad & \pi_{j,t} = P^Q_{j,t} Q_{j,t} - P^M_{j,t} M_{j,t}, \\ \text{s.t.} \quad & Q_{j,t} = D^B_j(P^Q_t, \delta_t; \eta), \\ & Q_{j,t} = F(K_{j,t}, L_{j,t}, M_{j,t}; \beta)\Omega_{j,t}. \end{aligned} \tag{9}$$

The two constraints create a chain linking the firm's price choice to all other variables. For a given price $P^Q_{j,t}$, the demand function determines $Q_{j,t}$, and the production function then pins down the required level of intermediate inputs $M_{j,t}$. Capital $K_{j,t}$ and labor $L_{j,t}$ are predetermined (Assumption 3), so intermediate inputs are the only input that adjusts. Both $Q_{j,t}$ and $M_{j,t}$ are, therefore, implicit functions of the price, with the intermediate input responding to price changes according to

$$\frac{\partial M_{j,t}}{\partial P^Q_{j,t}} = \frac{\frac{\partial D^B_j}{\partial P^Q_{j,t}}}{\frac{\partial F}{\partial M_{j,t}}\Omega_{j,t}}. \tag{10}$$

The numerator captures how demand responds to the firm's own price change, while the denominator transmits the resulting quantity change into the required intermediate inputs adjustment scaled by productivity. Differentiating profits with respect to $P^Q_{j,t}$ and substituting this expression in yields the first order condition when firms compete in prices:

$$\frac{\partial \pi_{j,t}}{\partial P^Q_{j,t}} = Q_{j,t} + P^Q_{j,t}\frac{\partial D^B_j}{\partial P^Q_{j,t}} - \frac{P^M_{j,t}}{\frac{\partial F}{\partial M_{j,t}}\Omega_{j,t}}\frac{\partial D^B_j}{\partial P^Q_{j,t}} = 0, \tag{11}$$

where $\frac{\partial D_j^B}{\partial P_{j,t}^Q}$ denotes the partial derivative of firm j's demand with respect to its own price, holding all competitors' prices fixed. Under most standard demand specifications, the profit function is strictly concave in the firm's own price, in these cases this FOC is both necessary and sufficient for a maximum.

This FOC, like its Cournot counterpart in equation (7), holds generally under any differentiable production function and demand system. The structural difference between the two cases lies in the relevant demand object: under Cournot competition, the firm's perceived marginal revenue depends on the slope of inverse demand $\frac{\partial D^C}{\partial Q_{j,t}}$ evaluated at competitors' quantities, whereas under Bertrand competition it depends on the own-price slope of demand $\frac{\partial D_j^B}{\partial P_{j,t}^Q}$ evaluated at competitors' prices.

As in the Cournot case, the FOC involves latent output quantities $Q_{j,t}$ and productivity $\Omega_{j,t}$, making it generally impossible to solve for $\Omega_{j,t}$ in closed form. The demand function maps the full vector of prices to individual firm quantities, so each firm's FOC depends on its own price and, through the production function constraint, on its own latent productivity and output quantity. I can apply the same MPEC approach using the FOC as a constraint, with the solver jointly recovering the vectors of latent productivities and output quantities that simultaneously satisfy all firms' optimality conditions while minimizing the GMM objective.

The structure of unobserved variables, constraints, and moment conditions is almost exactly the same as in the Cournot case. For each firm in each period, the same four latent variables $(Q_{j,t}, \Omega_{j,t}, \epsilon_{j,t}, \xi_{j,t})$ plus one demand shock δ_t per period need to be recovered, and the same structural equations, the relationship between observed and unobserved outputs (3), the production function (Assumption 1), the productivity development process (Assumption 2), the demand function, and the Bertrand FOC, provide the necessary constraints. The moment conditions follow from the same information set assumptions as in Section 2.1. The orthogonality of $\xi_{j,t}$ to fixed inputs, lagged inputs, and lagged productivity levels holds. The orthogonality of $\epsilon_{j,t}$ to all input choices can be used. The demand parameters can be identified through the exogeneity of input prices to demand shocks (Assumption 6).

The full MPEC estimator under Bertrand competition is then

$$
\begin{aligned}
\underset{\theta,\sigma}{\arg\min} \quad & \frac{1}{J \cdot T} A(X;\theta)^T W A(X;\theta) \\
s.t. \quad & Q_{j,t} + \frac{\partial D_j^B}{\partial P_{j,t}^Q} \left(P_{j,t}^Q - \frac{P_{j,t}^M}{\frac{\partial F}{\partial M_{j,t}} \Omega_{j,t}} \right) = 0, \\
& \omega_{j,t} = g(\omega_{j,t-1};\rho) + \xi_{j,t}, \\
& Q_{j,t} = F(K_{j,t}, L_{j,t}, M_{j,t};\beta)\Omega_{j,t}, \\
& Y_{j,t} = Q_{j,t} e^{\epsilon_{j,t}}, \\
& Q_{j,t} = D_j^B(P_t^Q, \delta_t;\eta), \qquad \forall t = 1, ..., T, \\
& \qquad\qquad\qquad\qquad\quad \forall j = 1, ..., J.
\end{aligned}
\tag{12}
$$

The MPEC differs from its Cournot counterpart in equation (8) in two parts. First, the FOC constraint encodes the Bertrand pricing condition rather than the Cournot quantity-setting condition, with the own-price demand partial derivative replacing the inverse demand partial derivative. Second, the demand constraint specifies firm-level demand D_j^B as a function of the full price vector P_t^Q rather than an inverse demand function of the quantity vector Q_t. Despite these differences, the estimation general approach of the GMM objective, the weighting matrix, and the moment conditions, remains identical across both competition modes. The choice between Cournot and Bertrand reduces to specifying the

appropriate FOC and demand structure.

3 Demand Specification: Latent vs. Observed Output

The MPEC estimators derived in Section 2 under assumptions 1-8a (or 8b) assume that demand is a function of the latent output quantity $Q_{j,t}$ and that firms choose $Q_{j,t}$ to maximize profits directly. Under this specification, the error term $\epsilon_{j,t}$ in the relationship between observed and unobserved output ($Y_{j,t} = Q_{j,t}e^{\epsilon_{j,t}}$) (Equation 3) is a pure measurement error that the firm neither observes nor anticipates. In this setting it does not enter the firm's decision problem at all. This is a common assumption in the production function estimation literature because it simplifies the first order conditions or the input demand functions used to invert for unobserved productivity $\Omega_{j,t}$.

However, similar to Gandhi et al. (2020), I can relax this assumption in two ways. First, even though the firm does not observe the realization of $\epsilon_{j,t}$ before making its input and output decisions, it may anticipate the distribution of the error term when choosing its inputs and output. This modification alone does not affect the firm's behavior if demand is a function of the latent output $Q_{j,t}$, because the error enters only after all quantities and prices are determined. Second, and more consequentially, demand may be a function of the observed output $Y_{j,t}$ rather than the latent output $Q_{j,t}$. This would mean that the firm actually produces $Y_{j,t}$, and buyers observe and respond to the quantities actually delivered to the market, not to the firm's intended production plan. When both modifications are combined, the firm anticipates the error distribution and demand depends on $Y_{j,t}$, the firm's problem becomes an expected profit maximization. The error term $\epsilon_{j,t}$ now enters the revenue function through demand, and the firm needs to take it into account when choosing intermediate inputs and output. In the Cournot case, this alters the FOC to

$$\frac{\partial Q_{j,t}}{\partial M_{j,t}} \left\{ E_\epsilon \left[\frac{\partial D^C}{\partial Y_{j,t}} e^{2\epsilon_{j,t}} \right] Q_{j,t} + E_\epsilon \left[D^C(Y_t, \delta_t; \eta) \cdot e^{\epsilon_{j,t}} \right] \right\} - P^M_{j,t} = 0 \tag{13}$$

Under Bertrand competition, the same logic applies. The firm chooses $M_{j,t}$ and hence $Q_{j,t}$ before observing $\epsilon_{j,t}$. After the error realizes, the observed output $Y_{j,t} = Q_{j,t}e^{\epsilon_{j,t}}$ enters the market. In the Bertrand setting, the firm sets its price $P^Y_{j,t}$ knowing that $\epsilon_{j,t}$ will realize and affect the quantity actually delivered to the market. Since demand depends on $Y_{j,t}$, the realized price adjusts through inverse demand $P^Y_{j,t} = (D^B_j)^{-1}(Y_{j,t}; P^Y_{-j,t}, \delta_t, \eta)$, so that revenue $P^Y_{j,t} \cdot Y_{j,t}$ is random. The firm therefore chooses its inputs to maximize expected revenue, taking $\epsilon_{j,t}$ into account. The modified FOC with respect to $M_{j,t}$ is

$$\frac{\partial Q_{j,t}}{\partial M_{j,t}} \left\{ E_\epsilon \left[\frac{\partial P^Y_{j,t}}{\partial Y_{j,t}} e^{2\epsilon_{j,t}} \right] Q_{j,t} + E_\epsilon \left[P^Y_{j,t} \cdot e^{\epsilon_{j,t}} \right] \right\} - P^M_{j,t} = 0, \tag{14}$$

where $\frac{\partial P^Y_{j,t}}{\partial Y_{j,t}} = \left(\frac{\partial D^B_j}{\partial P^Y_{j,t}} \right)^{-1}$ is the slope of firm j's inverse demand, holding competitors' prices fixed, evaluated at $Y_{j,t} = Q_{j,t}e^{\epsilon_{j,t}}$. This FOC has the same structure as its Cournot counterpart in Equation (13). Both contain expectations of exponential functions of $\epsilon_{j,t}$ multiplied by demand derivatives and price terms. The only difference is the strategic assumption embedded in the demand derivative, mirroring the distinction already present in the baseline FOCs of Section 2.

Both modified FOCs contain expectations over nonlinear functions of $\epsilon_{j,t}$, which in general depend on the full distribution of the error term. However, under the assumption that $\epsilon_{j,t}$ is homoscedastic, these expectations reduce to scalar constants that do not vary across firms or time periods. This

simplifies the FOCs considerably, as the expectations can be absorbed into two additional parameters to be estimated. The FOC for the Cournot case becomes

$$\frac{\partial Q_{j,t}}{\partial M_{j,t}} \left\{ \frac{\partial D^C}{\partial Y_{j,t}} \mu_2 Q_{j,t} + D^C(Y_t, \delta_t; \eta)\mu_1 \right\} = P^M_{j,t}, \tag{15}$$

and for the Bertrand case,

$$\frac{\partial Q_{j,t}}{\partial M_{j,t}} \left\{ \left(\frac{\partial D^B_j}{\partial P^Y_{j,t}} \right)^{-1} \mu_2 Q_{j,t} + P^Y_{j,t}\mu_1 \right\} = P^M_{j,t}, \tag{16}$$

with $\mu_1 = E[e^{\epsilon_{j,t}}]$ and $\mu_2 = E[e^{2\epsilon_{j,t}}]$.

The number of additional parameters can be reduced further by imposing distributional assumptions on $\epsilon_{j,t}$. If $\epsilon_{j,t}$ is normally distributed with variance σ^2_ϵ, the moment generating function of the normal distribution implies $\mu_1 = E[e^{\epsilon_{j,t}}] = e^{\sigma^2_\epsilon/2}$ and $\mu_2 = E[e^{2\epsilon_{j,t}}] = e^{2\sigma^2_\epsilon}$. Since $e^{2\sigma^2_\epsilon} = (e^{\sigma^2_\epsilon/2})^4 = \mu_1^4$, the two parameters collapse into one and both FOCs can be expressed with only one additional parameter μ_1. In this case, only one additional parameter needs to be estimated beyond those in the baseline specification.

While assuming demand to be a function of observed output $Y_{j,t}$ introduces the additional parameter(s) μ_1 and μ_2, it offers a substantial computational advantage by breaking the simultaneity between demand and production estimation. In the baseline MPEC of Section 2, demand and production function parameters must be estimated jointly because demand is a function of the unobserved output $Q_{j,t}$, which is itself determined by the production function parameters and unobserved productivity. This joint estimation creates a high-dimensional nonlinear constrained optimization problem in which the solver must simultaneously recover all latent variables and all structural parameters. When demand instead depends on the observed output $Y_{j,t}$, all variables entering the demand function (output and prices) are directly available in the data. The econometrician can, therefore, estimate the demand parameters in a first stage, since only the demand shock δ_t is unobserved. In a second stage, the production function parameters can be estimated via the MPEC-GMM estimator, taking the first-stage demand estimates as given. This two-stage procedure reduces the dimensionality of the constrained optimization problem, both in terms of the number of parameters and the number of unknown variables that the solver must recover simultaneously. This can be particularly valuable as the FOC constraints can be highly nonlinear functions of the latent variables and parameters, and reducing problem dimensionality can improve the performance of the numerical solvers substantially.

The switch from $Q_{j,t}$ to $Y_{j,t}$ in the demand function also has implications for the validity of the moment conditions used in estimation. The key orthogonality conditions from Section 2 remain valid. The FOC now involves expectations over $\epsilon_{j,t}$, but it does not change the timing. Since all fixed choices are made before $\epsilon_{j,t}$ is realized, the conditions based on $\epsilon_{j,t}$ that include fixed inputs or lagged input choices continue to hold regardless of whether the firm anticipates the distribution of $\epsilon_{j,t}$. Similarly, the orthogonality of $\xi_{j,t}$ to predetermined variables is unaffected, as the productivity process (Assumption 2) is independent of the specification of $\epsilon_{j,t}$. The moment conditions for demand estimation are also preserved. Input prices remain exogenous to both the demand shock δ_t and the error $\epsilon_{j,t}$ under Assumption 6. However, the interpretation of $\epsilon_{j,t}$ does change. In the baseline specification, $\epsilon_{j,t}$ is a pure measurement error with no effect on firms' choices. When the firm anticipates its distribution, $\epsilon_{j,t}$ becomes part of the economic environment. Its variance affects the firm's optimal input and output

choices through the modified FOC. This means that, in principle, the distribution of $\epsilon_{j,t}$ is identified not only from the residual variation in the observation equation but also from the firm's behavioral response to output uncertainty, captured by the parameters μ_1 and μ_2 in the FOC. However, when using the two-stage procedure described above, the second-stage's standard errors need to account for the estimation uncertainty propagated from the first stage. Therefore, analytical adjustments or bootstrap procedures that take the variance of the first-stage estimator into account, need to be applied to recover valid variance estimates.

4 Relationship to the Literature

The recent production function estimation literature is built around the proxy variable approach of Olley and Pakes (1996), Levinsohn and Petrin (2003), and Ackerberg et al. (2015), which uses an observable input, such as investment or intermediate inputs, to proxy for unobserved productivity and address the resulting simultaneity bias. A large subsequent literature has extended this framework in various directions: Gandhi et al. (2020) and Bond and Söderbom (2005) analyze identification failures for gross-output specifications; Doraszelski and Jaumandreu (2018), Demirer (2020), Raval (2019), and Zhang (2019) accommodate non-Hicks-neutral productivity; and De Loecker and Warzynski (2012); De Loecker et al. (2020) embed the estimator in models with variable markups. An alternative strand, including Griliches and Ringstad (1971), Doraszelski and Jaumandreu (2013), Grieco et al. (2016), Gandhi et al. (2020), and Trunschke and Judd (2024), uses firms' first-order conditions directly as identifying restrictions rather than relying on a nonparametric or analytical proxy inversion.

All of the above approaches share the common structure, that they proxy for unobserved productivity by inverting the firm's intermediate input demand function or FOCs, relying on the scalar unobservability condition that input demand is a monotone function of productivity alone. Under perfect competition this condition is often satisfied, but under imperfect competition, the intermediate input demand also depends on the market demand shock and competitors' actions, violating scalar unobservability and invalidating the first-stage inversion. Doraszelski and Li (2025) formalize this failure and show that applying the standard proxy variable estimator when invertibility fails yields structurally biased estimates. Beyond this diagnostic result, they propose modifications to the standard control function approach that reduce the bias without abandoning the proxy variable framework by augmenting the first-stage with a bias correction and modifying the second stage's moments to improve its robustness towards first-stage estimation noise.

Ackerberg and De Loecker (2024) propose a solution within the control function framework by exploiting the equilibrium structure in specific cases of imperfect competition. In a homogeneous-good Cournot market, they demonstrate that competitors' aggregate output $Q_{-j,t}$ is a sufficient statistic and productivity can be recovered from the augmented inversion of intermediate inputs demand. They extend this argument to logit and nested-logit Nash-Bertrand models of differentiated-product markets, where aggregate market quantities play the same role. Jaumandreu (2025) takes a complementary approach and shows that under general conditions the ratio of the intermediate input expenditure share to the short-run elasticity of scale provides a consistent proxy for the inverse markup-adjusted marginal cost, yielding a linear cost-share estimator that is robust to arbitrary forms of market power. Jaumandreu further derives a test for the competitive assumption or for functional form misspecification. Brand (2020) shows that production functions can be identified using a nonclassical measurement error approach that does not rely on specifying demand.

5 Simulation: Homogeneous Goods Cournot

In this section, I evaluate the finite-sample performance of the MPEC-GMM estimator derived in Section 2 using simulated data. The simulation serves two purposes. First, to verify that the estimator can recover the true structural parameters when the data generating process satisfies all model assumptions, and second, to demonstrate the practical feasibility of the approach. I specialize the general framework to a homogeneous goods Cournot market (Assumption 8a) with linear inverse demand, which provides a basic setting that retains the key identification challenges discussed in Section 2.1. Even in this simple setting, FOC has multiple roots and, therefore, cannot be analytically inverted for $\Omega_{j,t}$ as in the perfect competition case in Trunschke and Judd (2024). The inverse demand function is

$$P_t^Q = \eta_1 - \eta_2 \sum_j Q_{j,t} + \delta_t. \tag{17}$$

Firms produce output $Q_{j,t}$ using a Cobb-Douglas production function,

$$Q_{j,t} = \beta_0 K_{j,t}^{\beta_K} L_{j,t}^{\beta_L} M_{j,t}^{\beta_M} \Omega_{j,t}. \tag{18}$$

For simplicity, productivity development process (Assumption 2) is specified as a linear AR(1),

$$\omega_{j,t} = \rho \omega_{j,t-1} + \xi_{j,t}. \tag{19}$$

Substituting these functional forms into the general Cournot profit maximization problem of equation (6) yields the following constrained static optimization problem for each firm j in period t, where the firm chooses $M_{j,t}$ and $Q_{j,t}$ taking competitors' quantities, its own capital $K_{j,t}$ and labor $L_{j,t}$, and the input price $P_{j,t}^M$ as given:

$$\begin{aligned} \max_{M_{j,t}, Q_{j,t}, P_{j,t}^Q} \quad & \pi_{j,t} = P_{j,t}^Q Q_{j,t} - P_{j,t}^M M_{j,t} \\ s.t. \ \text{I:} \quad & P_t^Q = \eta_1 - \eta_2 \sum_j Q_{j,t} + \delta_t, \\ \text{II:} \quad & Q_{j,t} = \beta_0 K_{j,t}^{\beta_k} L_{j,t}^{\beta_l} M_{j,t}^{\beta_m} \Omega_{j,t}. \end{aligned} \tag{20}$$

Differentiating profits with respect to $M_{j,t}$, applying the chain rule through the production function, and using the linearity of inverse demand yields the FOC:

$$\frac{\partial \pi(\cdot)}{\partial M_{j,t}} = \beta_m P_t^Q Q_{j,t} - \eta_2 \beta_m Q_{j,t}^2 - M_{j,t} P_{j,t}^M = 0. \tag{21}$$

The parameter vector $\theta = (\beta_0, \beta_k, \beta_l, \beta_m, \rho, \eta_1, \eta_2)$ contains seven unknowns: four production function parameters, one persistence parameter from the productivity process, and two demand parameters. I construct ten moment conditions, each derived from the orthogonality restrictions described in Section 2.1.

For the demand parameters, I exploit the zero-mean assumption on the demand shock δ_t and its orthogonality to input prices under Assumption 6. Specifically, $E[\delta_t] = 0$ provides a first moment, and $E[\delta_t \cdot \sum_j P_{j,t}^M] = 0$ provides a second, where input prices serve as instruments because they affect output decisions through the profit maximization problem but are exogenous to demand shocks.

For the measurement error $\epsilon_{j,t}$, the zero-mean assumption $E[\epsilon_{j,t}] = 0$ and the orthogonality of $\epsilon_{j,t}$

to intermediate inputs $E[\log(M_{j,t}) \cdot \epsilon_{j,t}] = 0$, which holds because $M_{j,t}$ is chosen before $\epsilon_{j,t}$ is realized (Assumption 4), provide two further moments.

The productivity shock $\xi_{j,t}$ generates the largest block of moments. Its zero mean assumption $E[\xi_{j,t}] = 0$ provides one moment. Because capital and labor are predetermined with respect to $\xi_{j,t}$ (Assumption 3), both their contemporaneous levels and their lags are valid instruments: $E[\log(K_{j,t}) \cdot \xi_{j,t}] = 0$, $E[\log(L_{j,t}) \cdot \xi_{j,t}] = 0$, $E[\log(K_{j,t-1}) \cdot \xi_{j,t}] = 0$, and $E[\log(L_{j,t-1}) \cdot \xi_{j,t}] = 0$. Similarly, Assumption 2 implies $E[\log(\Omega_{j,t-1}) \cdot \xi_{j,t}] = 0$, adding another moment. Finally, the ergodic distribution assumption yields $E[\log(\Omega_{j,t})] = 0$, which helps identifying the production function's scale parameter β_0. The sample analogue of these ten moment conditions is

$$A(X, \sigma; \theta) = \begin{pmatrix} \frac{1}{J \cdot T} \sum_{t=1}^{T} \sum_{j=1}^{J} \delta_t \\ \frac{1}{J \cdot T} \sum_{t=1}^{T} \sum_{j=1}^{J} \delta_t (\sum_i P_{j,t}^M) \\ \frac{1}{J \cdot T} \sum_{t=1}^{T} \sum_{j=1}^{J} \xi_{j,t} \\ \frac{1}{J \cdot T} \sum_{t=1}^{T} \sum_{j=1}^{J} \log(M_{j,t}) \epsilon_{j,t} \\ \frac{1}{J \cdot T} \sum_{t=1}^{T} \sum_{j=1}^{J} \log(K_{j,t}) \xi_{j,t} \\ \frac{1}{J \cdot T} \sum_{t=1}^{T} \sum_{j=1}^{J} \log(L_{j,t}) \xi_{j,t} \\ \frac{1}{J \cdot T} \sum_{t=1}^{T} \sum_{j=1}^{J} \log(K_{j,t-1}) \xi_{j,t} \\ \frac{1}{J \cdot T} \sum_{t=1}^{T} \sum_{j=1}^{J} \log(L_{j,t-1}) \xi_{j,t} \\ \frac{1}{J \cdot T} \sum_{t=1}^{T} \sum_{j=1}^{J} \log(\Omega_{j,t-1}) \xi_{j,t} \\ \frac{1}{J \cdot T} \sum_{t=1}^{T} \sum_{j=1}^{J} \log(\Omega_{j,t}) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}. \tag{22}$$

The system is overidentified with ten moments for seven parameters, providing three degrees of overidentification. For the baseline estimation, I use the identity matrix as the weighting matrix $W = I$. While a two-step efficient GMM estimator using the optimal weighting matrix $W = \hat{S}^{-1}$, where $\hat{S}$ is the estimated variance-covariance matrix of the moment conditions, could improve asymptotic efficiency. However, this requires $\hat{S}$ to be well-conditioned, which may be difficult to ensure in small samples.

Combining the moment conditions with the five structural equality constraints, the FOC, the production function, the productivity development process, the relationship between $Q_{j,t}$ and $Y_{j,t}$, and the inverse demand function, yields the following MPEC-GMM estimator for the homogeneous goods

Cournot case with linear inverse demand:

$$
\begin{aligned}
\underset{\theta,\sigma}{\arg\min} \quad & (J \cdot T) A(X;\theta)^T W A(X;\theta) \\
s.t. \quad & P_{j,t}^M M_{j,t} = \beta_m P_t^Q Q_{j,t} - \eta_2 \beta_m Q_{j,t}^2, \\
& \xi_{j,t} = \log(\Omega_{j,t}) - \rho_\omega \log(\Omega_{j,t-1}), \\
& Q_{j,t} = \beta_0 K_{j,t}^{\beta_k} L_{j,t}^{\beta_l} M_{j,t}^{\beta_m} \Omega_{j,t} \\
& \epsilon_{j,t} = \log\left(\frac{Y_{j,t}}{Q_{j,t}}\right), \\
& P_t^Q = \eta_1 - \eta_2 \sum_j Q_{j,t} + \delta_t, \qquad \begin{array}{l} \forall t = 1, ..., T, \\ \forall j = 1, ..., J. \end{array}
\end{aligned} \tag{23}
$$

As discussed in Section 2.1, the FOC under linear demand is quadratic in the unobserved output quantity $Q_{j,t}$, yielding two positive roots that both satisfy the second order conditions. While the remaining constraints, the production function, the observation equation, and the productivity development process, each have a unique solution for the latent variables given the parameters, the multiplicity introduced by the FOC means that the full system of constraints has more than one solution for a given parameter vector θ. This is the situation where the MPEC formulation demonstrates its advantage. The solver does not need to select the correct root ex ante. Instead, the GMM objective function, evaluated at the moment conditions, discriminates between the multiple solutions. Only one combination of parameters and latent variables simultaneously satisfies all constraints and minimizes the criterion function, because the moment conditions impose additional restrictions that the incorrect root generically violates. The MPEC-GMM estimator, therefore, identifies all parameters and latent quantities despite the non-uniqueness of the constraint system.

I simulate a panel of $J = 20$ firms over $T = 20$ periods from a data generating process that satisfies all model assumptions, and apply the MPEC-GMM estimator to assess how well it recovers the true parameter values. in the simulated data. This is a deliberately small sample, both in cross-section and time series.

Table 5 reports the estimation results. Overall, the estimator recovers the true parameter values well even in this small sample, with all 95% confidence intervals covering the true values. The precision varies across parameters. The output elasticities, which are the central objects of interest, are recovered with varying degrees of precision. The intermediate input elasticity β_m is estimated most accurately, with a relative error of only 0.63% and a substantially smaller standard error than the other production function parameters. This high precision is expected: β_m is directly pinned down by the FOC constraint. The labor elasticity β_l is estimated with a relative error of 6.62% and a confidence interval that covers the true value. Labor is identified through the moment conditions exploiting its orthogonality to $\xi_{j,t}$, but as a fixed input chosen before the productivity shock, the identifying variation comes from cross-sectional differences in labor choices conditional on expected productivity, which provides less information per observation than the FOC. The capital elasticity β_k exhibits a larger relative bias of 34.45%. Its standard error is correspondingly large, so the confidence interval covers the true value. The scale parameter β_0 has the largest relative bias at 35.2%. The productivity development process parameter ρ is recovered with a relative error of 3.21% and a standard error of 0.05. The demand parameters η_1 and η_2 are both estimated with relative errors below 0.1%.

Table 1: Estimation Results

Parameter	Truth	Estimate	Std. Err	t-Stat	p-Value	95% CI		Abs. Error (%)
β_0	1.0000	1.3052	0.483	2.7023	0.0069	0.3585	2.2518	0.3052
β_k	0.2000	0.1311	0.062	2.113	0.0346	0.0095	0.2526	0.3445
β_l	0.5000	0.4669	0.0921	5.0705	0.0	0.2864	0.6474	0.0662
β_m	0.3000	0.3019	0.0207	14.5517	0.0	0.2612	0.3426	0.0063
ρ	0.7000	0.6775	0.0515	13.1576	0.0	0.5766	0.7785	0.0321
η_1	1000.0	1000.9	9.0395	110.727	0.0	983.2	1018.6	0.0009
η_2	3.0000	3.0016	0.0396	75.7051	0.0	2.9239	3.0793	0.0005

Notes: Asymptotic standard errors calculation for GMM with MPEC is described in detail in Appendix A. Results are based on 400 observations (20 time periods x 20 firms). Burn-in period is 10 years.

6 Conclusion

Imperfect competition and strategic interactions pose key issues to standard production function estimators leading to biased estimates. This paper develops an MPEC-GMM estimator for production functions in markets with imperfect competition and strategic interactions. The central idea is to retain all latent variables, firm-level productivities $\Omega_{j,t}$, unobserved output quantities $Q_{j,t}$, productivity innovations $\xi_{j,t}$, measurement errors $\epsilon_{j,t}$, and demand shocks δ_t, as explicit optimization variables and to enforce the firm's first order condition, the production function, the productivity development process, the relationship between observed output $Y_{j,t}$, unobserved output $Q_{j,t}$ and the error term $\epsilon_{j,t}$, and the demand system simultaneously as equality constraints. The approach is general enough to be combined with more flexible demand specifications such as the Almost Ideal Demand System (Deaton and Muellbauer, 1980), which provides a second-order approximation to any demand system without requiring data on product characteristics. This approach sidesteps the invertibility failures that arise under imperfect competition when applying standard control function methods, because no analytical or nonparametric inversion of the FOC or input demand function is required. The estimator builds on the constrained optimization framework of Trunschke and Judd (2024) and extends it from perfect competition to oligopolistic settings where firms face downward-sloping demand and condition their decisions on competitors' actions.

I derive the estimator under two canonical modes of competition: under Cournot and Bertrand competition. Despite the different strategic variables and demand objects, the estimation structure is common in both settings: the same five types of constraints, the same latent variables, and the same moment conditions apply, with only the FOC and demand specification changing.

Identification relies on standard information-set assumptions. The productivity shock $\xi_{j,t}$ is orthogonal to all predetermined variables, capital, labor, their lags, and lagged productivity, because these are chosen before $\xi_{j,t}$ is realized. The measurement error $\epsilon_{j,t}$ is orthogonal to all input choices because it is realized after the firm's decisions. Input prices, which are exogenous to demand shocks under the price-taking assumption in input markets, serve as instruments for the demand parameters. The ergodic distribution of productivity provides an additional moment that helps identify the scale parameter of the production function.

An important specification choice is whether demand depends on latent output $Q_{j,t}$ or observed output $Y_{j,t}$ (Section 3). Under the baseline Q-specification, $\epsilon_{j,t}$ is a pure measurement error that does not enter the firm's decision problem, and all structural parameters must be estimated jointly. Under the Y-specification, firms anticipate the distribution of $\epsilon_{j,t}$ and maximize expected profits, introducing

two additional parameters into the FOC, which collapse to a single parameter under normality of the error term. However, this brings a computational advantage because all variables entering the demand function are observed, demand can be estimated in a first stage and the second-stage MPEC only needs to consider a reduced number of parameters. Key moment conditions remain valid under both specifications, as the timing of input choices relative to $\epsilon_{j,t}$ is unchanged.

A Monte Carlo simulation for the case of a Cobb-Douglas production function with linear inverse demand under homogenous goods Cournot competition confirms the estimator's ability to recover all structural parameters in a small panel. The output elasticities are recovered well: the intermediate input elasticity β_m is estimated with the highest precision, while the capital elasticity β_k is less precisely estimated. The demand parameters are recovered with relative errors below 0.1%, and all 95% confidence intervals cover the true values. The simulation also demonstrates that the MPEC formulation successfully handles the non-uniqueness of the FOC, which is quadratic in $Q_{j,t}$ under linear demand, because the moment conditions discriminate between the multiple roots of the constraint system.

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A Asymptotic Variance Estimator for GMM-MPEC

This appendix derives the asymptotic variance estimator for the MPEC-GMM estimator introduced in Section 2. Because the MPEC formulation treats structural parameters $\theta \in \mathbb{R}^p$ and latent variables $\sigma \in \mathbb{R}^m$ as joint decision variables linked by constraints $h(\theta, \sigma) = 0$, recovering the correct Jacobian of the moment conditions requires an implicit-function-theorem (IFT) correction. In the estimator of this paper, $\theta = (\beta_0, \beta_K, \beta_L, \beta_M, \rho, \eta_1, \eta_2)'$ collects the production function and demand parameters, while $\sigma = (\{\Omega_{j,t}\}, \{Q_{j,t}\}, \{\delta_t\})'$ collects the latent productivity levels, output quantities, and demand shocks. The constraints $h(\theta, \sigma) = 0$ encode the production function, the first-order conditions, and the demand equation as described in Section 2.

The MPEC-GMM estimator $(\hat{\theta}, \hat{\sigma})$ solves

$$(\hat{\theta}, \hat{\sigma}) \in \underset{(\theta,\sigma)\in\Theta\times\Sigma}{\arg\min}\; g_N(\theta, \sigma)'\, W_N\, g_N(\theta, \sigma) \qquad \text{s.t.} \quad h(\theta, \sigma) = 0, \tag{A1}$$

where $g_N(\theta, \sigma) = \frac{1}{N}\sum_{i=1}^{N} g(\theta, \sigma; X_i)$ is the k-dimensional sample moment vector built from the moment conditions in Section 2, and $W_N \xrightarrow{p} W_0$ is a positive-definite weight matrix. As shown by Su and Judd (2012), the MPEC estimator is numerically equivalent to the standard nested fixed-point GMM estimator and inherits its asymptotic properties. Under standard regularity conditions, $\hat{\theta}$ is consistent and asymptotically normal with

$$\sqrt{N}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}\big(0,\; (G_0' W_0 G_0)^{-1}\, G_0' W_0\, S_0\, W_0 G_0\, (G_0' W_0 G_0)^{-1}\big)\,, \tag{A2}$$

where $G_0 = \frac{d}{d\theta'}\mathbb{E}[g(\theta, \sigma(\theta); X_i)]\big|_{\theta_0}$ is the $k \times p$ Jacobian of the population moments evaluated along the feasible manifold $h(\theta, \sigma(\theta)) = 0$, $S_0 = \mathbb{E}[g_i g_i']$ is the variance of the moment vector at the truth, and W_0 is the probability limit of the weight matrix. When $W_0 = S_0^{-1}$ (efficient GMM), the sandwich simplifies to $(G_0' S_0^{-1} G_0)^{-1}$.

Estimating W_0 and S_0 is straightforward: W_0 is the matrix supplied to the criterion, and S_0 is consistently estimated by $\hat{S} = \frac{1}{N}\sum_{i=1}^{N} \hat{g}_i\, \hat{g}_i'$ with $\hat{g}_i = g(\hat{\theta}, \hat{\sigma}; X_i)$. Recovering G_0, however, requires care. In the MPEC formulation the solver treats θ and σ as independent decision variables, so its automatic differentiation delivers partial Jacobians

$$G^\theta = \left.\frac{\partial g}{\partial \theta'}\right|_{(\hat{\theta},\hat{\sigma})}, \quad G^\sigma = \left.\frac{\partial g}{\partial \sigma'}\right|_{(\hat{\theta},\hat{\sigma})}, \tag{A3}$$

that each hold the other block fixed. Neither alone is a consistent estimator of G_0 because both ignore the endogenous response of σ to changes in θ imposed by the constraints.

To correct for this, note that $h(\theta, \sigma(\theta)) = 0$ holds identically along the feasible manifold. Total differentiation with respect to θ gives

$$\frac{d\, h(\theta, \sigma(\theta))}{d\theta'} = \frac{\partial h}{\partial \theta'} + \frac{\partial h}{\partial \sigma'}\,\frac{\partial \sigma}{\partial \theta'} = 0, \tag{A4}$$

so, defining $H^\theta = \partial h / \partial \theta'$ and $H^\sigma = \partial h / \partial \sigma'$, the implicit function theorem yields

$$\frac{\partial \sigma}{\partial \theta'} = -(H^\sigma)^{-1}\, H^\theta, \tag{A5}$$

provided H^σ is invertible. I implement the MPEC estimator of this paper in the mathematical mod-

elling language JuMP (Lubin et al., 2023) built on top of the Julia programming language (Bezanson et al., 2017) and solve the optimization problem with Artelys Knitro (Byrd et al., 2006). JuMP supplies exact derivatives to the solver via automatic differentiation, and I use JuMP's automatic differentiation together with ForwardDiff.jl (Revels et al., 2016) to obtain the partial Jacobians G^θ, G^σ, H^θ, and H^σ at the solution. In the estimator of this paper, the constraints link each $(\Omega_{j,t}, Q_{j,t}, \delta_t)$ tuple to the structural parameters through nonlinear equations, making H^σ a large but sparse matrix. I therefore compute the IFT correction via a sparse linear solve rather than explicit inversion. The chain rule then delivers the corrected Jacobian when evaluated at $(\hat{\theta}, \hat{\sigma})$:

$$\hat{G} = G^\theta - G^\sigma \left(H^\sigma\right)^{-1} H^\theta. \tag{A6}$$

Substituting $\hat{G}$, $\hat{S}$, and W_N into the sandwich formula (A2) and dividing by N yields the estimated variance-covariance matrix of $\hat{\theta}$.